

Bioenergetics and electron carriers

Thermodynamics and oxidation-reduction reactions

Supportive material



Bioenergetics and electron carriers

The ability to obtain, store and use energy is characteristic of living organisms. Bioenergetics is the study of how organisms obtain and use energy.

The energy transformations that take place in living cells obey the same laws of thermodynamics that govern all natural processes. The basic principles of thermodynamics are:

1. First law of thermodynamics or law of conservation of energy: in any physical or chemical change the total energy in an isolated system remains constant.
2. Second law of thermodynamics: any physical or chemical change proceeds spontaneously in such a way that the entropy increases.

Living organisms maintain their internal organization by exchanging matter and energy with their surroundings. Although entropy may decrease locally within an organism, the total entropy of the organism and its environment increases.

Watch this video as an introduction to the relationship between life and energy

Spontaneity of biological processes: Gibbs free energy

A fundamental aspect in the study of metabolic processes is the ability to determine in which direction they will take place spontaneously.

Suppose a reaction in which compounds A and B give rise to compounds C and D:

$$A + B \rightleftharpoons C + D$$

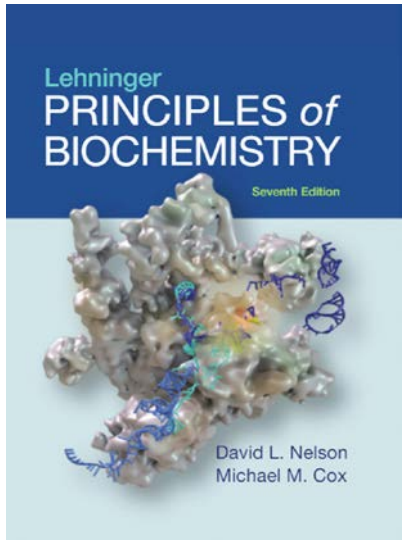
It is important to determine whether the process will progress towards the formation of C and D, or towards the formation of A and B, as well as whether a change in the physiological conditions of the cell, such as an increase in compound C, can change the direction of the process.

The second law of thermodynamics determines that processes take place spontaneously in the direction in which the entropy increases. However, entropy changes are not easily measurable and it is required to know both the entropy change of the reaction and the environment.

In the case of biological processes, the Gibbs free energy change (ΔG) is the parameter used to determine in which direction a reaction will proceed.

Additional study material

ADI document: Bioenergetics and electron carriers



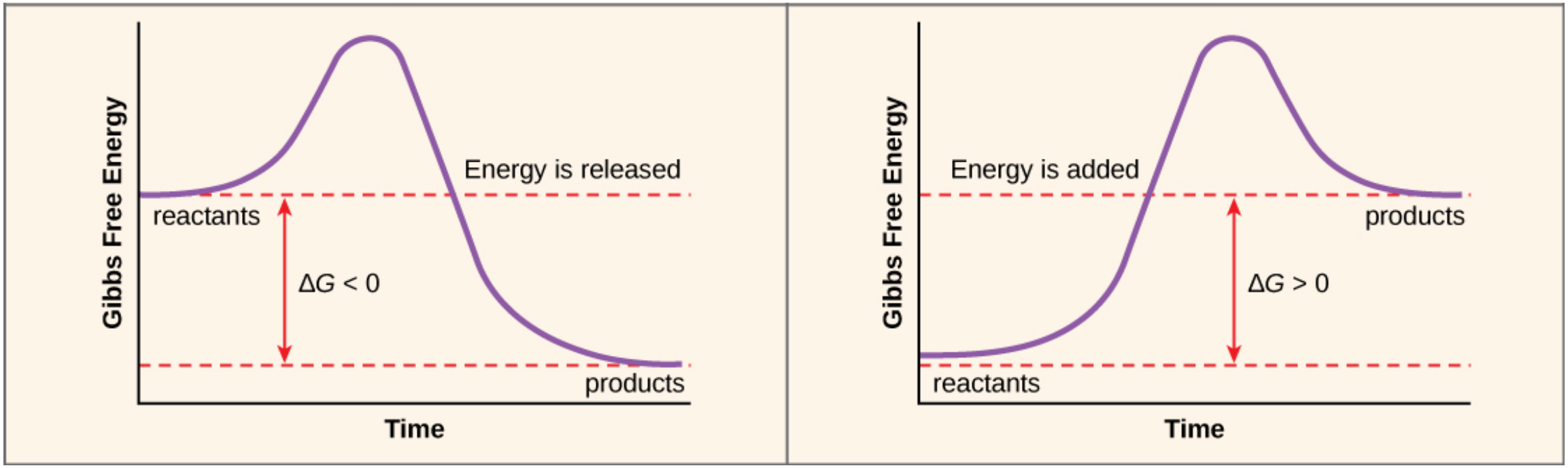
Reading

Lehninger’s PoB Chapter 13: Bioenergetics

Learning goals

- Basic concepts of thermodynamics applied to biology
- Concept and implications of free-energy change
- Concept of reversible and irreversible reactions
- Oxidation-reduction reactions in biology
- Oxidation states of carbon
- Description and role of electron carriers

Gibbs free energy change (ΔG)

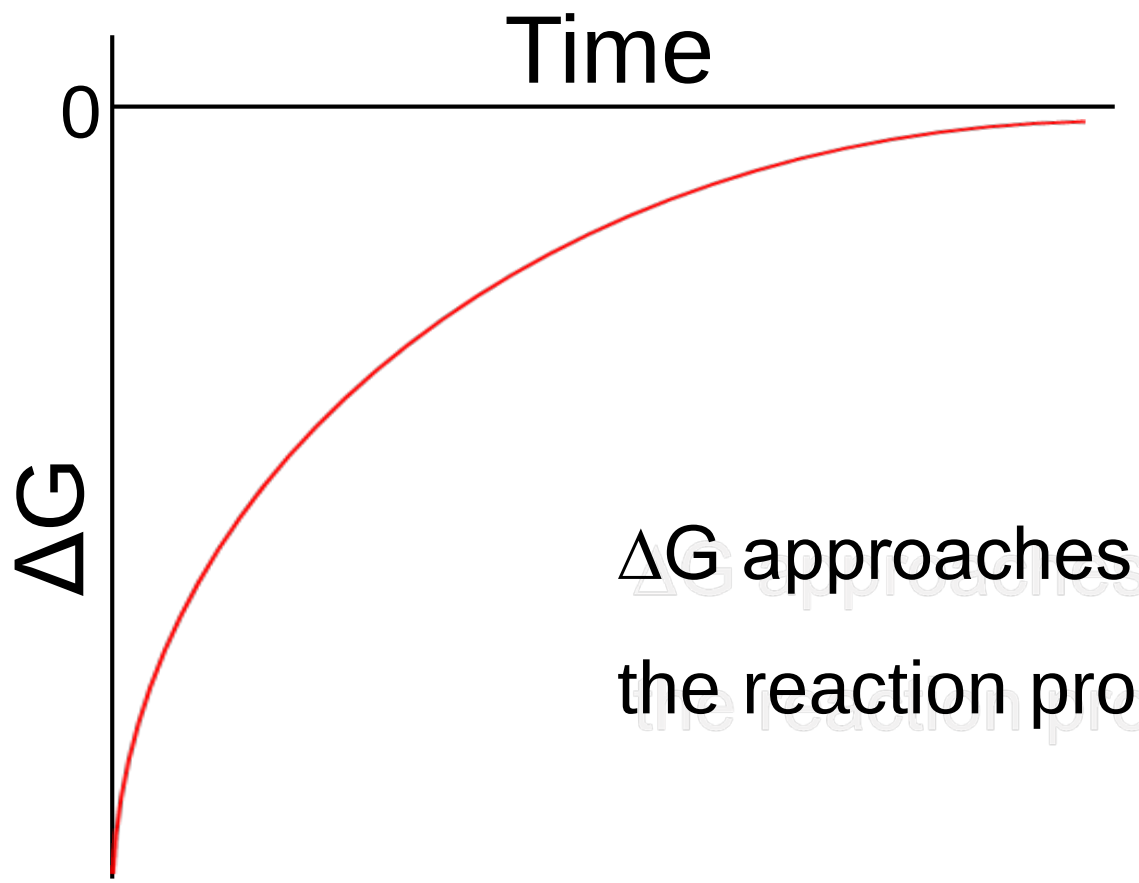


If ΔG is < 0 then the reaction is exergonic and proceeds in the forward direction

If ΔG is > 0 then the reaction is endergonic and proceeds in the reverse direction

If ΔG is = 0 then the reaction is at equilibrium

Gibbs free energy change



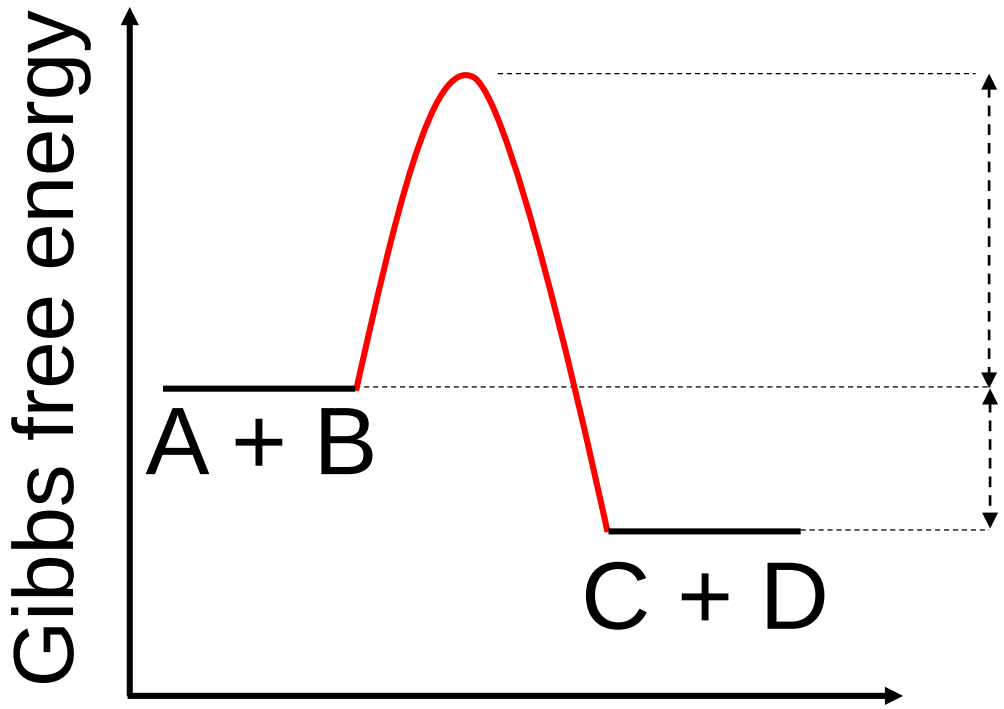
ΔG approaches zero as the reaction progresses

TABLE 13-4 Standard Free-Energy Changes of Some Chemical Reactions

Reaction type	$\Delta G'^{\circ}$ (kJ/mol)	$\Delta G'^{\circ}$ (kcal/mol)
Hydrolysis reactions		
Acid anhydrides		
Acetic anhydride + H_2O → 2 acetate	–91.1	–21.8
ATP + H_2O → ADP + P_i	–30.5	–7.3
ATP + H_2O → AMP + PP_i	–45.6	–10.9
PP_i + H_2O → 2P_i	–19.2	–4.6
UDP-glucose + H_2O → UMP + glucose 1-phosphate	–43.0	–10.3
Esters		
Ethyl acetate + H_2O → ethanol + acetate	–19.6	–4.7
Glucose 6-phosphate + H_2O → glucose + P_i	–13.8	–3.3
Amides and peptides		
Glutamine + H_2O → glutamate + NH_4^+	–14.2	–3.4
Glycylglycine + H_2O → 2 glycine	–9.2	–2.2
Glycosides		
Maltose + H_2O → 2 glucose	–15.5	–3.7
Lactose + H_2O → glucose + galactose	–15.9	–3.8
Rearrangements		
Glucose 1-phosphate → glucose 6-phosphate	–7.3	–1.7
Fructose 6-phosphate → glucose 6-phosphate	–1.7	–0.4
Elimination of water		
Malate → fumarate + H_2O	3.1	0.8
Oxidations with molecular oxygen		
Glucose + 6O_2 → 6CO_2 + 6H_2O	–2,840	–686
Palmitate + 23O_2 → 16CO_2 + 16H_2O	–9,770	–2,338

Lehninger 7th ed.

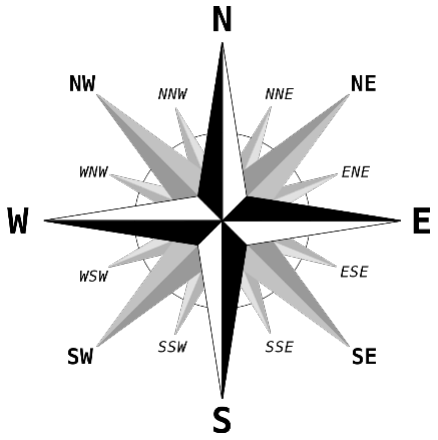
Free-energy profile of a reaction



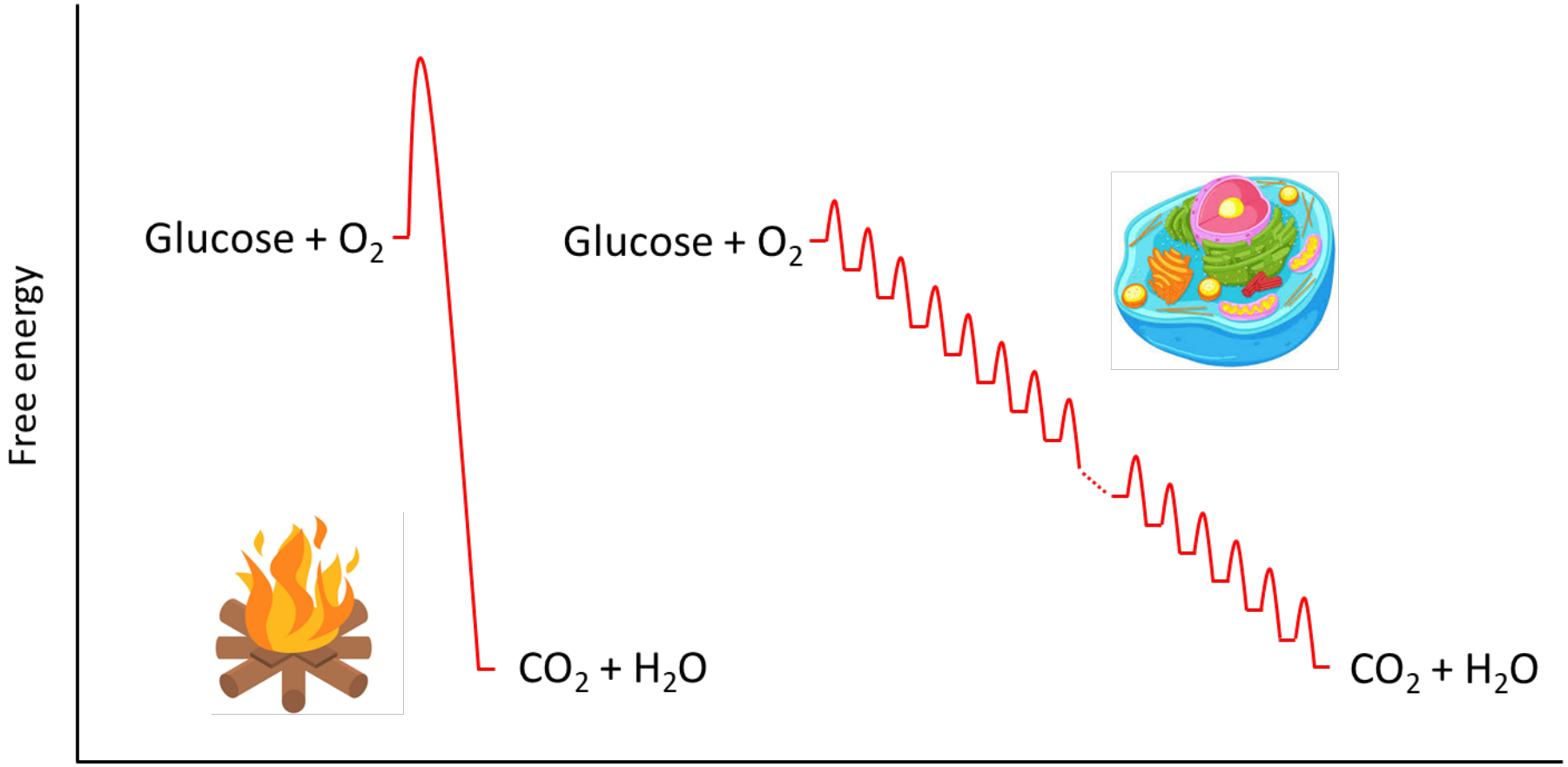
Activation energy ($\Delta G^{\ddagger}$)



Free-energy change (ΔG)



ΔG is independent of the pathway



$$\text{Glucose} + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O \qquad \Delta G^{o\prime} = -2874\ \text{kJ/mol}$$

ΔG of a reaction

The free-energy change of a reaction depends on:

- The standard free energy change
- The concentrations of reactants and products

$$A + B \rightleftharpoons C + D$$

$$\Delta G = \Delta G^{o'} + RT \ln \frac{[C][D]}{[A][B]}$$

$\Delta G^{o'}$: standard free-energy change (at pH 7)

R: gas constant (8.314 $J \cdot mol^{-1} \cdot K^{-1}$)

T: absolute temperature (K)

Reversible vs. irreversible reactions

In metabolism, a reaction is considered reversible when physiologically achievable changes in metabolite concentrations can reverse its direction. A reaction is considered irreversible when the conditions required to reverse it are incompatible with normal cellular conditions

Glucose 6P $\rightleftharpoons$ fructose 6P $\quad \Delta G^{o'} = +1.7$ kJ/mol

Phosphoenolpyruvate + ADP $\longrightarrow$ pyruvate + ATP $\quad \Delta G^{o'} = -31.4$ kJ/mol

Metabolic pathways are exergonic

Glycolysis (Glucose → pyruvate)

$$\text{Glucose} + 2\ \text{ADP} + 2\ \text{Pi} + 2\ \text{NAD}^+ \rightarrow$$

$$2\ \text{pyruvate} + 2\ \text{ATP} + 2\ \text{NADH} + 2\ \text{H}^+ + 2\ \text{H}_2\text{O}$$

$\Delta G^{o\prime} = -51$ kJ/mol

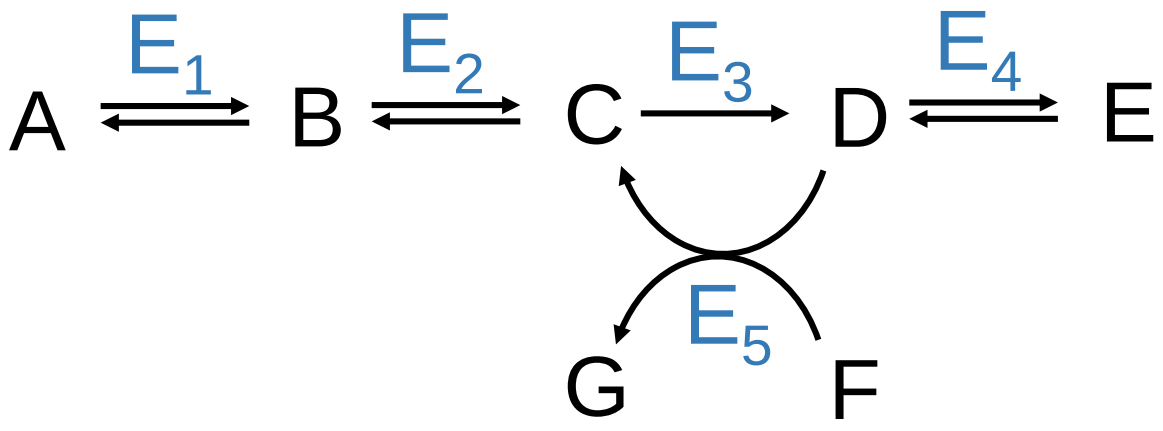
Gluconeogenesis (Pyruvate → glucose)

$$2\ \text{Pyruvate} + 4\ \text{ATP} + 2\ \text{GTP} + 2\ \text{NADH} + 2\ \text{H}^+ + 4\ \text{H}_2\text{O} \rightarrow$$

$$\text{glucose} + 4\ \text{ADP} + 2\ \text{GDP} + 6\ \text{P}_i + 2\ \text{NAD}^+$$

$\Delta G^{o\prime} = -38$ kJ/mol

Metabolic pathways are exergonic



$A \longrightarrow E \qquad \Delta G < 0$

$E + F \longrightarrow A + G \qquad \Delta G < 0$

- The degradation process of a biomolecule is different from its biosynthetic process
- This allows a coordinated and reciprocal regulation

Reaction coupling

Glucose + P_i ⟶ Glucose 6-phosphate + H_2O $\Delta G^{o'}$ = +13.8 kJ/mol

ATP + H_2O ⟶ ADP + P_i $\Delta G^{o'}$ = -30.5 kJ/mol

Glucose + ATP ⟶ Glucose 6-phosphate + ADP

$\Delta G^{o'}$ = +13.8 + (- 30.5) = - 16.7 kJ/mol

Phosphoryl transfer potential

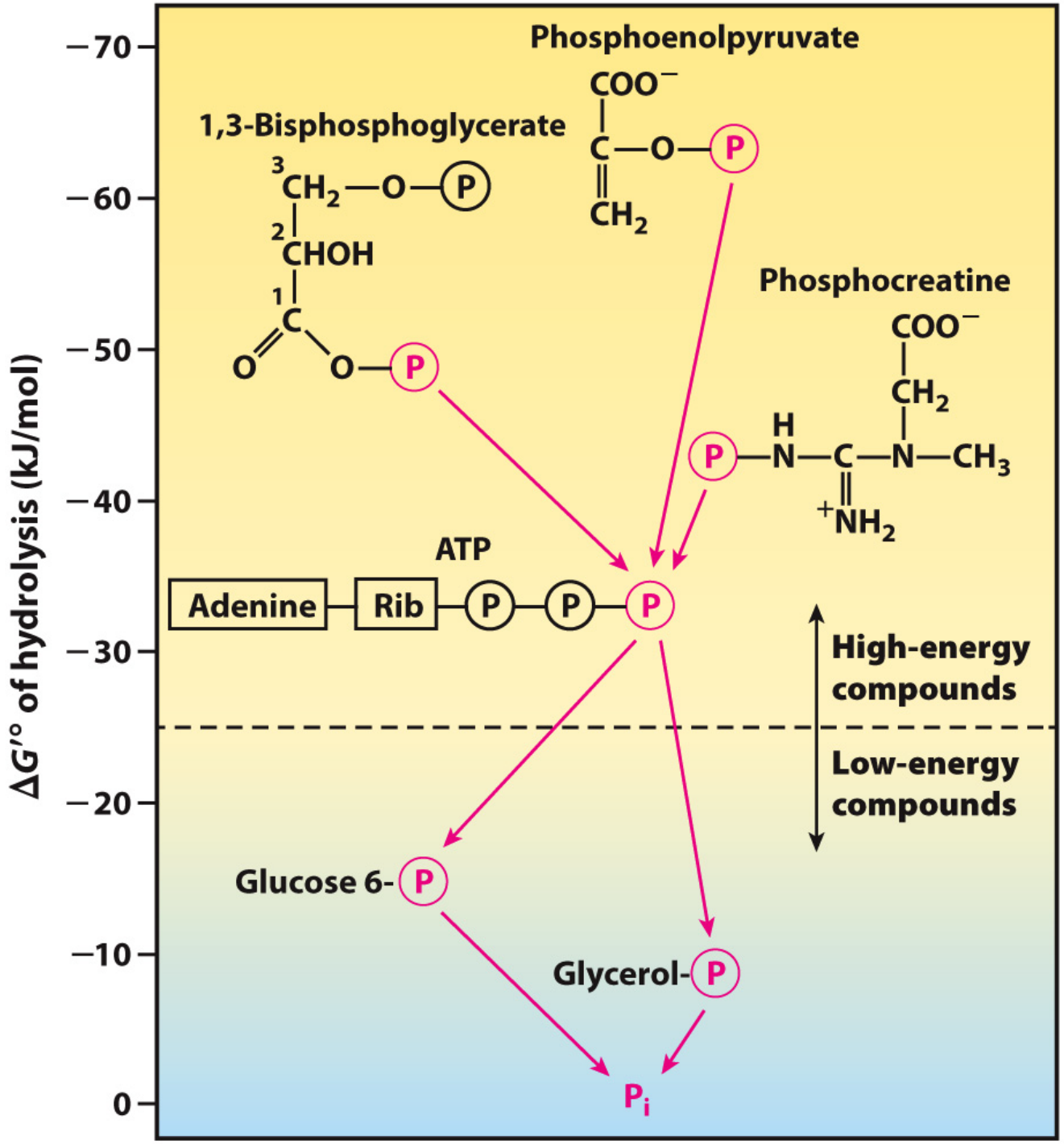


Figure 13-19
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CATABOLISM

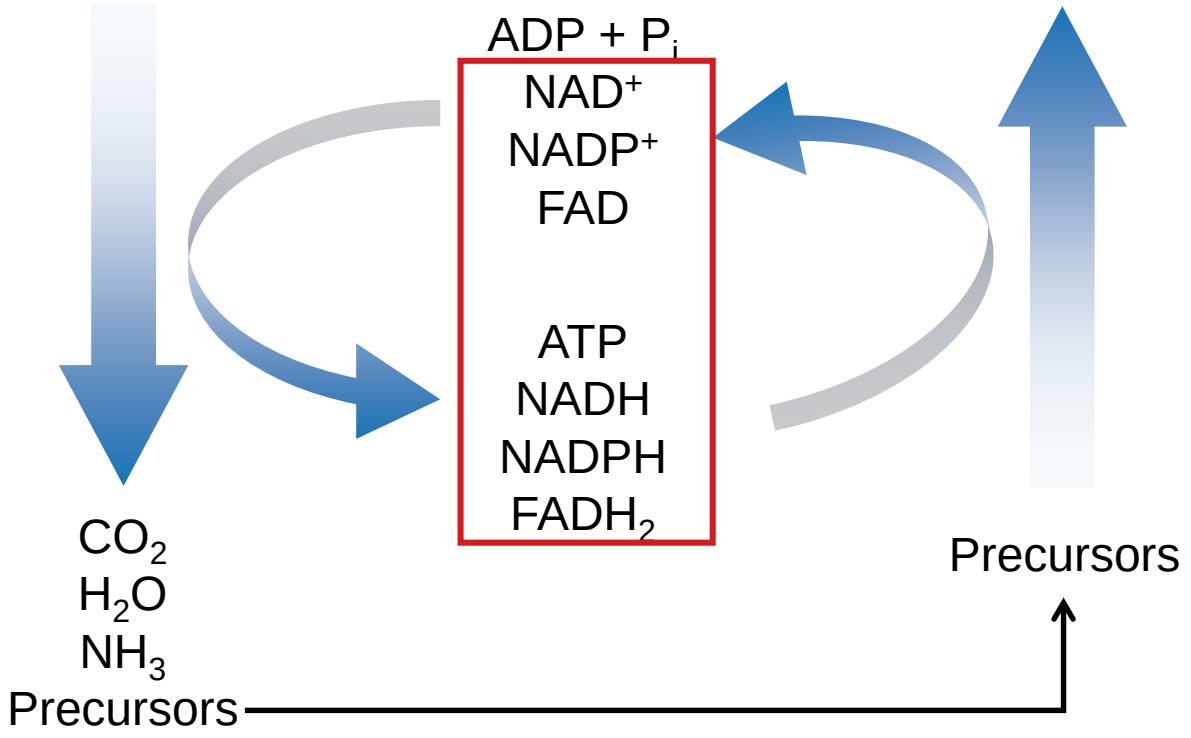
Energy-containing nutrients

Carbohydrates
Lipids
Proteins

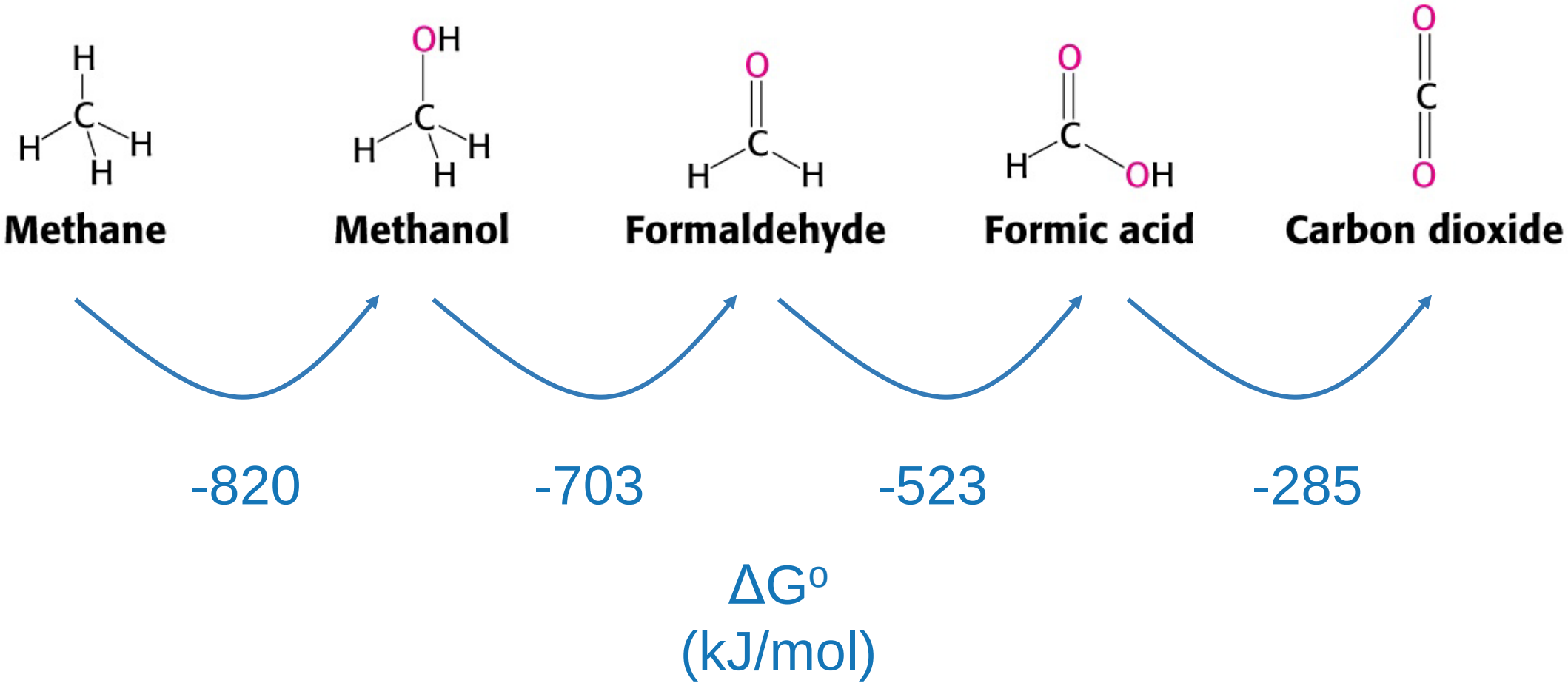
ANABOLISM

Macromolecules

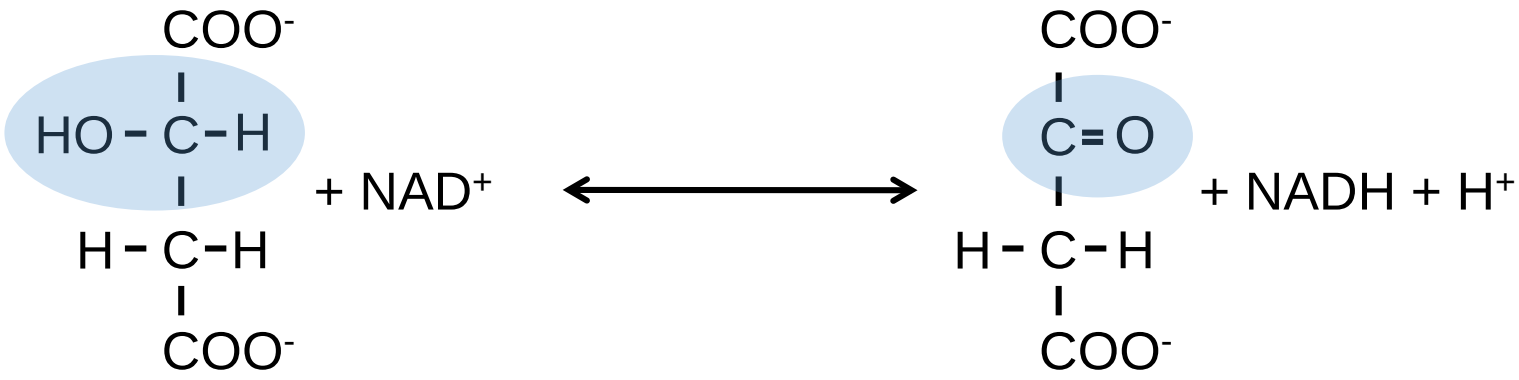
Carbohydrates
Lipids
Proteins
Nucleic acids



Carbon releases free energy upon oxidation



Oxidation-reduction reactions



Malate Oxaloacetate

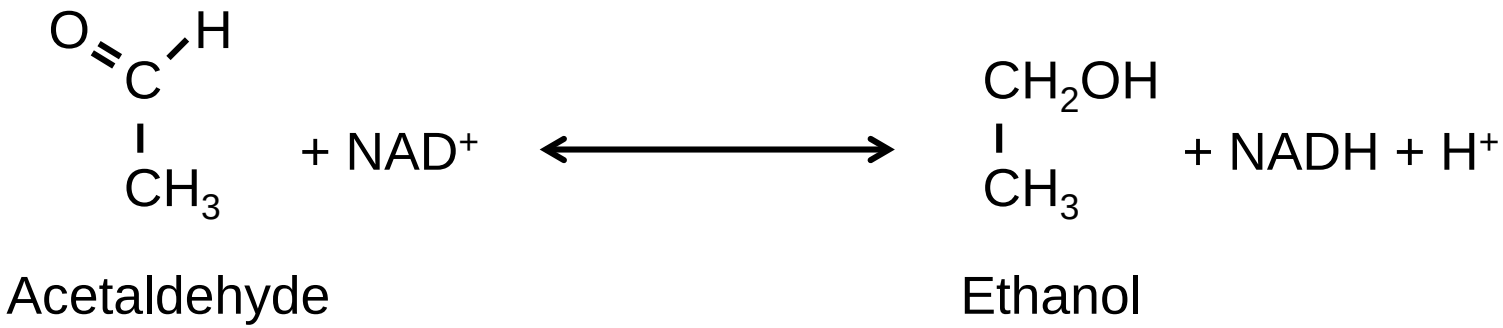
Half-reaction 1 $\text{Malate} \longleftrightarrow \text{Oxaloacetate} + 2e^-$

Half-reaction 2 $NAD^+ + 2e^- \longleftrightarrow NADH + H^+$

Malate loses electrons → it is oxidized

NAD^+ gains electrons → it is reduced

What is wrong with this reaction?

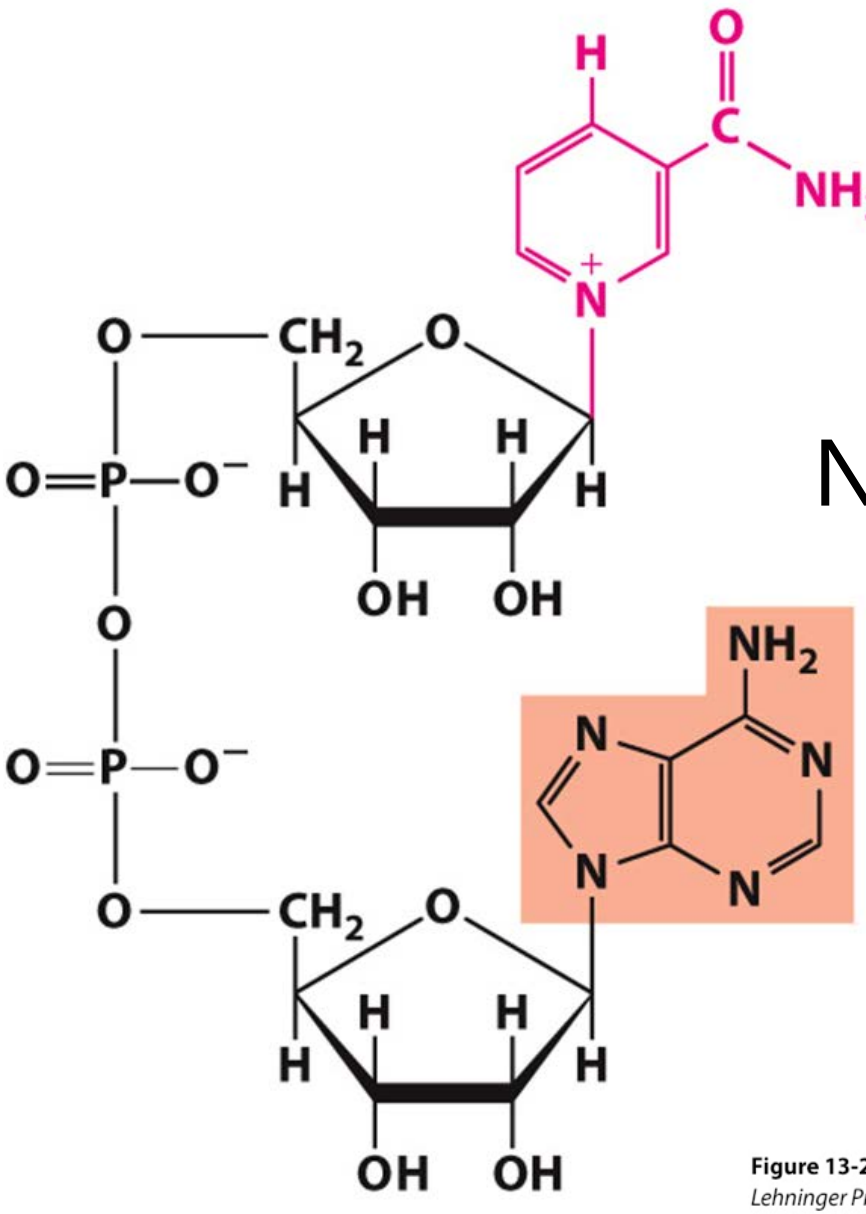


Electron carriers in metabolism

AH_2 + oxidized carrier ↔ A + reduced carrier

NAD^+	NADH
$NADP^+$	NADPH
FAD	$FADH_2$
FMN	$FMNH_2$

NAD^+ and $NADP^+$



$$NAD(P)^+ + 2H^+ + 2e^- \leftrightarrow NAD(P)H + H^+$$

Figure 13-24a
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NAD^+ and $NADP^+$

- They are commonly called pyridine nucleotides
- The nicotinamide ring (active group) is derived from niacin (vitamin B_3)
- They dissociate from the enzyme after the reaction

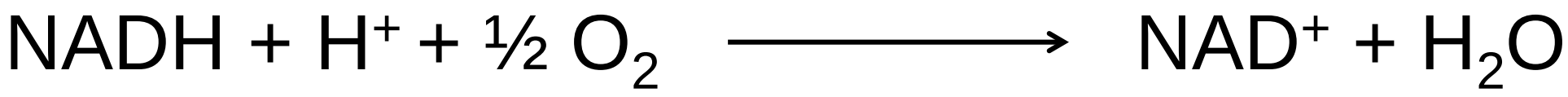
$$AH_2 + NAD(P)^+ \leftrightarrow A + NAD(P)H + H^+$$

NAD^+ and $NADP^+$

Enzyme	*Coenzyme*
Isocitrate dehydrogenase	NAD^+
α-Ketoglutarate dehydrogenase	NAD^+
Glucose 6-phosphate dehydrogenase	$NADP^+$
Malate dehydrogenase	NAD^+
Glutamate dehydrogenase	NAD^+ or $NADP^+$
Glyceraldehyde 3-phosphate dehydrogenase	NAD^+
Lactate dehydrogenase	NAD^+
Alcohol dehydrogenase	NAD^+

Lehninger 4th ed.

Oxygen is the terminal electron acceptor in aerobic metabolism



$\Delta G^{o'} = -220$ kJ/mol

FAD and FMN

$$FAD + 2H^+ + 2e^- \leftrightarrow FADH_2$$

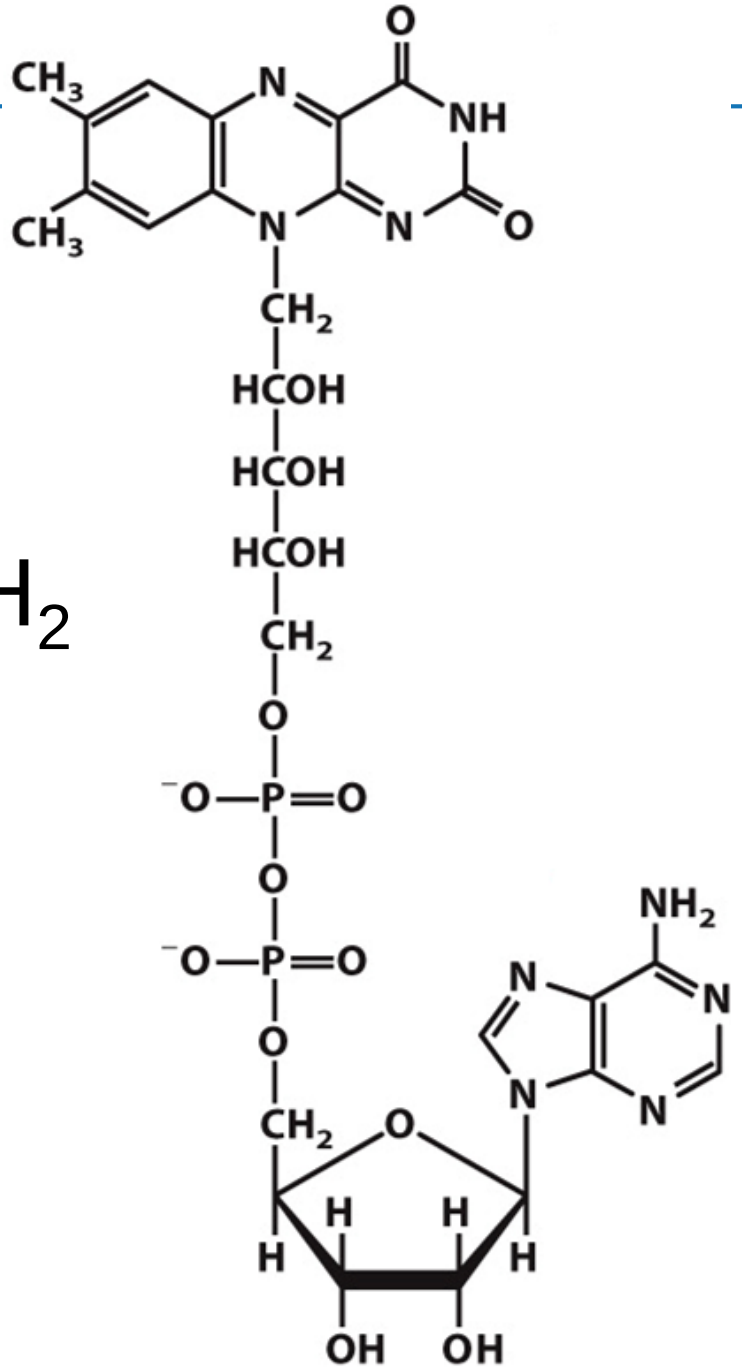


Figure 13-27
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FAD and FMN

- They are commonly called flavin nucleotides
- Their active group is the isoalloxazine ring, which is derived from riboflavin (vitamin B_2)
- They are usually tightly bound to flavoproteins.
- A typical reaction in which they participate is the formation of double bond between carbon atoms

$$FAD + R\text{-}CH_2\text{-}CH_2\text{-}R' \leftrightarrow FADH_2 + R\text{-}CH{=}CH\text{-}R'$$

FAD and FMN

Enzyme	*Flavin nucleotide*
Acyl–CoA dehydrogenase	FAD
Dihydrolipoyl dehydrogenase	FAD
Succinate dehydrogenase	FAD
Glycerol 3-phosphate dehydrogenase	FAD
Thioredoxin reductase	FAD
NADH dehydrogenase (Complex I)	FMN
Glycolate oxidase	FMN

Lehninger 4th ed.